

Day 13 of training

DOCUMENT OBJECT MODEL (DOM)----> JavaScript Topic

The Document Object Model (DOM) is a programming interface for web documents.

It represents the page structure as a tree of objects, allowing JavaScript to access, modify, and update the content, structure, and style of a webpage dynamically.

In short:

DOM lets JavaScript interact with HTML and CSS to make web pages dynamic and interactive.

`document.getElementById()`

It is a JavaScript method used to select an HTML element by its id.

Syntax:

`document.getElementById("your-id")`

Key Points:

Always unique — id should be one per element.

Returns a single element.

Useful for accessing and modifying specific parts of a page.

`document.getElementsByClassName()`

It is a JavaScript method used to select all HTML elements with a specific class name.

Syntax:

`document.getElementsByClassName("your-class-name")`

Key Points:

Returns an `HTMLCollection` (like an array of elements).

You can use index to access specific elements.

You can loop through all elements with that class.

`let elements = document.getElementsByClassName("msg");`

```
elements[0].style.color = "green"; // changes first <p> text color
```

```
document.getElementsByTagName()
```

It is a JavaScript method used to select all HTML elements with a specific tag name (like p, div, h1, etc).

Syntax:

```
document.getElementsByTagName("tag-name")
```

Key Points:

Returns an HTMLCollection of all matching tags.

You can access elements using index or loop through them.

Common tags: p, div, img, li, button, etc.

```
document.querySelector()
```

It selects the first element that matches a CSS selector (like a class, id, tag, etc.).

Syntax:

```
document.querySelector("selector")
```

```
document.querySelectorAll()
```

It selects all elements that match a CSS selector, and returns a NodeList (like an array).

Syntax

```
document.querySelectorAll("selector")
```

Feature	`querySelector`	`querySelectorAll`
-----	-----	-----
Selects	First matching element	All matching elements
Return Type	Single element (Node)	NodeList (array-like)

| Use case | One element | Multiple elements |

how do you choose?

Use this simple rule:

◆ Use `getElementById()` — if you know the element has a unique id.

◆ Use `querySelector()` — when:

You need to select something using complex selectors like:

`.class`

`#id`

`div > p`

You only want the first match

◆ Use `querySelectorAll()` — when you want to apply something to multiple elements using a selector.

What is `createElement()`?

`createElement()` is a method in JavaScript used to create a new HTML element using code — before adding it to your webpage.

Syntax:

```
document.createElement("tagName");
```

What is `appendChild()` in JavaScript?

`appendChild()` is used to add an element inside another element — usually a newly created element.

Syntax:

```
parentElement.appendChild(childElement);
```

Things to Remember:

appendChild() can only add one node at a time.

It adds the child as the last child (at the bottom).

If you use it on an element that's already on the page, it moves it.

What is insertBefore() in JavaScript?

It's used to insert a new element before an existing one inside a parent.

Syntax

```
parent.insertBefore(newElement, existingElement);
```